

Software Architecture

SE6105

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(BSc Honours in Software Engineering)

SE6105 – Course Content

- Topic 01: Basic concepts & principles about software architecture
- Topic 02: Introduction to Software Architectural pattern
- Topic 03: ADL
- Topic 04: 4+1 Architecture
- Topic 05: Practical approaches & methods for Create & Analyse software architecture
- Topic 06: Quality attributes of software architectures
- Topic 07: Examples in architectural design applications & case studies in software architecture (N tier architecture, SOA, Cloud, etc.)

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Examples in architectural design applications & case studies

Why Study Real-World Architectures?

- Bridges theory → practice
- Helps in design decision-making
- Understand trade-offs
- Prepares for industry scenarios

Common Architectural Styles in Practice

- Layered (N-tier) Architecture
- Service-Oriented Architecture (SOA)
- Microservices Architecture
- Cloud-Native Architecture
- Event-Driven Architecture

Case Study Approach

When analyzing architecture:

- Identify system requirements
- Identify quality attributes
- Choose architecture style
- Evaluate trade-offs

Section 1: N-Tier Architecture

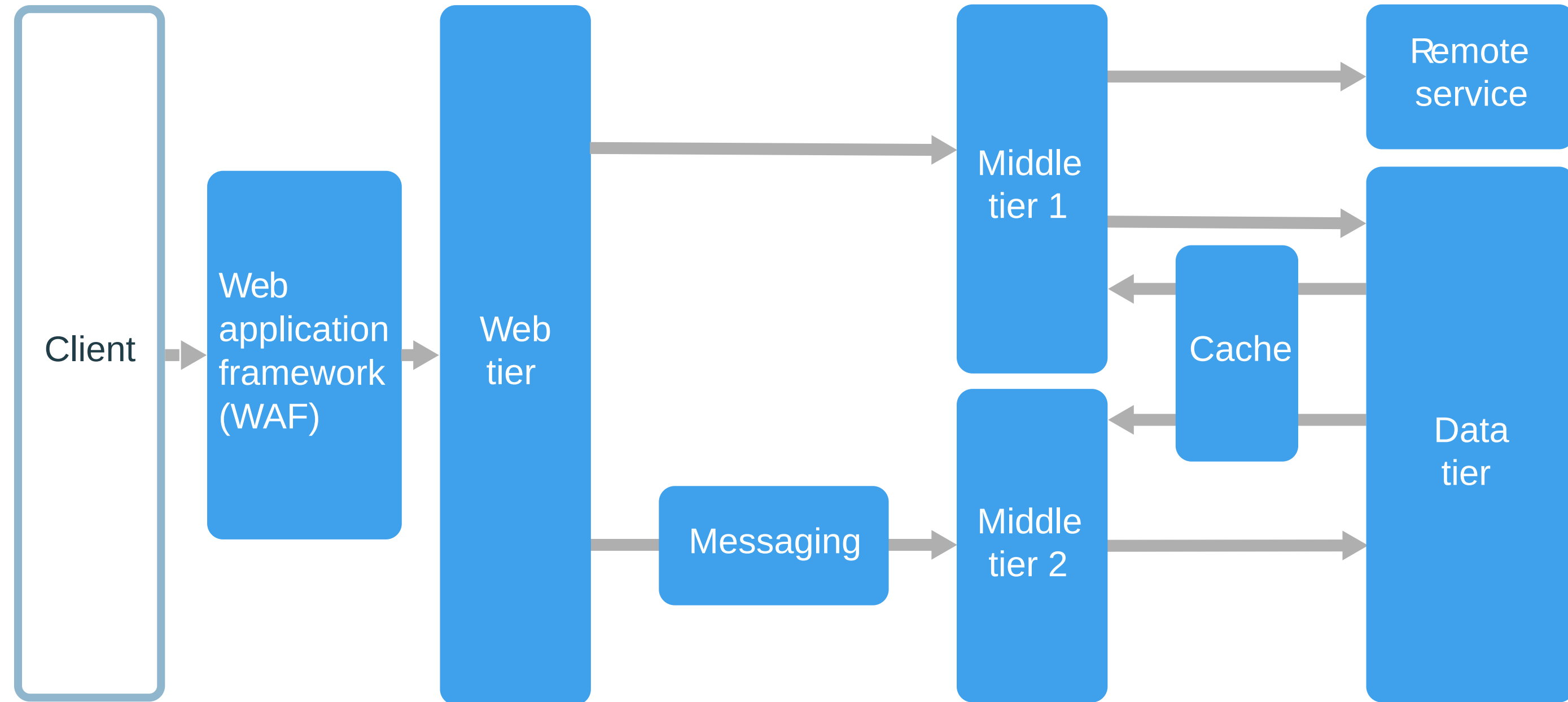
What is N-Tier Architecture?

- Separation into multiple layers

Common layers:

- Presentation
- Business Logic
- Data Access

N-Tier Architecture Diagram



Advantages of N-Tier

- Separation of concerns
- Easier maintenance
- Scalability
- Reusability

Limitations of N-Tier

- Performance overhead
- Tight coupling between layers
- Hard to scale independently

Case Study: E-Commerce System

- Frontend: Web/Mobile UI
- Backend: Business logic (orders, payments)
- Database: Product & user data

Why N-tier?

- Clear separation
- Maintainability

Section 2: Service-Oriented Architecture (SOA)

What is SOA?

- System composed of services
- Services communicate via protocols (HTTP, SOAP)

Key Characteristics of SOA

- Loose coupling
- Service reuse
- Interoperability

SOA Example

- Banking system:
- Account service
- Payment service
- Notification service

Advantages & Challenges of SOA

Advantages:

- Flexibility
- Reusability

Challenges:

- Complex management
- Service orchestration overhead

Microservices Architecture

What is Microservices?

- Small, independent services
- Each service handles one business capability

Microservices vs Monolithic

Feature	Monolithic	Microservices
Deployment	Single	Independent
Scalability	Limited	High
Complexity	Low	High

Case Study: Netflix-like System

- User service
- Recommendation service
- Streaming service

Benefits:

- High scalability
- Fault isolation

Challenges in Microservices

- Distributed system complexity
- Network latency
- Data consistency

Section 4: Cloud Architecture

What is Cloud Architecture?

- Applications hosted on cloud platforms

Uses services like:

- Compute
- Storage
- Networking

Cloud Architecture Example

Online learning platform:

- Frontend → Cloud hosting
- Backend → APIs
- Storage → Cloud database

Benefits of Cloud Architecture

- Scalability (auto-scaling)
- Cost efficiency
- High availability

Section 5: Event-Driven Architecture

What is Event-Driven Architecture?

- System reacts to events
- Components communicate asynchronously

Example: Ride-Sharing App

Event flow:

- User books ride → Event triggered
- Driver notified → Another event
- Payment processed

Advantages & Use Cases

Advantages:

- High scalability
- Loose coupling

Use Cases:

- Real-time systems
- IoT systems

Summary & Key Takeaways

- Different architectures suit different needs
- Trade-offs are unavoidable
- Always consider quality attributes:
 - Performance
 - Scalability
 - Maintainability

Examples in architectural design
applications & case studies

Thank You!